

gSender Android performance report

Build 55 update | 22 September 2026

Build 55 completes the selected suite on K50 and K90; Lenovo stops during the 40 MiB preview at the memory guard. Loading is slower than Build 53, and rendering issues remain. This report covers cumulative Android changes relative to official gSender 1.6.4. Measurements compare Android builds, not stock desktop performance.

Measured changes from Build 53 to Build 55

Tablet	Relief completion display delay s	Relief frame gap p95 ms	40 MiB load and preview s
Lenovo TB-8506F	0.55 to 0.58	50 to 67-83	42.59 to Stopped
Rebecca K50	0.55 to 0.58	50-66 to 66-83	17.29 to 19.80
Rebecca K90	0.57 to 0.56	150 to 150	17.68 to 19.84

Lower values are better. Frame gaps measure animation callbacks, not presented FPS or touch latency. Layout and viewport changes limit attribution to one code change. Build 53 had already reduced mountain completion-display delays from 24-69 seconds to about 0.56 seconds; that gain remains.

Cumulative improvements relative to live gSender

Android execution and connections. The desktop application is hosted in Android WebView with its existing backend and native USB bridge. The port moved from Node.js Mobile to a source-built Node 24.21.0 runtime, added USB recovery and SLB/Lite autoconnection, and retained scoped dev pendant/jogging code with Android touch and sensor fixes. These are portability and reliability changes, not measured desktop speed gains.

Less routine interface work. Toolpath geometry is retained while the position marker moves. Progress reuses line totals or counts incrementally. Hidden knob polling and disabled log formatting are avoided; release code and resources are optimized. A component test reduced 30 position-triggered geometry rebuilds to zero.

Lower job-storage overhead. The sender uses an 8-byte offset index per nonblank command instead of retaining individual line strings. A host-only 20 MiB microbenchmark reduced extra line storage from 26.1 to 4.4 MiB. RAM-aware budgeting replaces the port's early fixed heap limits; this ARM32 APK allows up to 1,536 MiB of V8 old-space, not total app memory.

Bounded and adaptive previews. Bounded worker bitmaps, a 1,600-pixel edge limit, reusable surfaces and transferable images reduce large-path overhead. Build 55 adds indexed density selection and clipped SVG for sparse detail. These runs show no overall speed gain and expose a clipping defect; the source fix is not in the APK.

Fresher status with less console traffic. Build 53 batches display copies of serial lines, bounds console history, pauses hidden-console rendering, and coalesces superseded position/progress snapshots. GrblHAL status polling changes from 250 to 100 ms. Alarms and state transitions retain ordering; command transmission and acknowledgements remain intact.

Repeatable diagnostics. Tools now includes offline job simulation, pan and 2× zoom workloads, checksum verification, memory/heartbeat logging, preview-stage errors and JSON export. These additions make regressions measurable on each tablet.

Device statistics for Build 55

Statistic	Lenovo TB-8506F	K50	K90
Android version	11	15	16
Android reported RAM GiB	1.77	3.70	8.00
Minimum available system RAM MiB	248	1,505	4,768
Peak host and Node PSS MiB	290	453	579
Maximum heartbeat response age ms	1579	474	460
Run duration minutes	19.11	19.28	19.26

RAM is Android-reported. Host PSS excludes the isolated WebView. Lenovo reached 248 MiB available while preparing the 40 MiB raster and triggered the 256 MiB benchmark stop. Its Build 53 run completed with a 335 MiB minimum. These samples do not identify which process caused the extra pressure.

Loading and preview completion in Build 55

Job	Lenovo seconds	K50 seconds	K90 seconds
Contour 128 KiB	1.22	0.85	0.77
Arcs 1 MiB	11.98	6.13	7.07
Real mountain relief 4.77 MiB	10.62	4.87	4.29
Dense relief 5 MiB	7.92	3.59	3.28
Dense relief 20 MiB	26.96	10.71	11.08
Dense relief 30 MiB	35.87	14.80	14.85
Dense relief 40 MiB	Stopped	19.80	19.84

Test coverage and development evidence

Each tablet verified 328,139 commands across three complete jobs with matching ordered checksums and zero receive overflows at about 500 commands/s. Larger files use 65-68-second streaming samples. Lenovo reached 30 MiB; K50 and K90 reached 40 MiB. Physical USB/CNC throughput and long endurance are not measured.

Earlier milestones: Builds 46-47 bounded previews after large-job failures; Build 48 rejected GPU rasterization after a Lenovo unresponsive-app event; Build 51 added bitmap caching; Builds 52-53 added onboard tests and reduced display backlog. Build 55 passed 331 automated tests and release checks before these device runs.

New findings and remaining work

All tablets logged two empty SVG frames during contour zoom, consistent with the reproduced visible-area clipping bug. Readiness alone missed it. Other job views stayed raster, so sparse-detail benefits remain untested. The source-only clipping/resize fix passed 18 focused tests and pixel checks; it has not run on tablets. Next, reduce peak preview memory and profile the persistent K90 rendering lag before claiming a performance gain.

One run per device. Layout/framing changed since Build 53. K50/K90 stayed portrait; Lenovo briefly rotated during the 20 MiB case, then returned. Memory samples can miss peaks and do not establish maximum file size. The simulator shares tablet resources. No new APK was built for this report.

Sources: [Official gSender 1.6.4 release](#); [MAXMAKE relief project](#). Local evidence: Build 53/55 tablet JSON exports, build55-comparison.md and build55-analysis.json; earlier build notes and MEMORY-PERFORMANCE.md. The relief fixture substitutes tool-change stops for simulation and is not a machine-qualified cutting file.